When Does Graph Structure Help Noisy-Label Intrusion Detection? A Leakage-Audited Study of Rare-Class Training-Signal Retention

Tianqi Guo[1*], Weiguo Wang[2], Jian Wang[2], Shiyin Zhu[2]

[1*]University of Science and Technology of China, No. 96 Jinzhai Road, Baohe District, Hefei, 230026, Anhui, P.R. China.
[2]New H3C Technologies Co., Ltd., P.R. China.

*Corresponding author(s). E-mail(s): guotianqi@mail.ustc.edu.cn;
Contributing authors: wwg@h3c.com; wang.jianE@h3c.com;
zhushiyin@h3c.com;

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

All three datasets are publicly available from their original providers: CIC-IDS-2017 from the Canadian Institute for Cybersecurity, University of New Brunswick; CESNET-TLS-Year22 from Zenodo (DOI: 10.5281/zenodo.10608607); and UNSW-NB15 from the University of New South Wales, Canberra. The corrected CICIDS-2017 labels used in the real-noise experiment are derived by applying the publicly released labelling rules of Liu et al. (CNS 2022, https://github.com/GintsEngelen/CNS2022_Code) to the original release. The derived artifacts required to reproduce our results—code, configuration files, frozen data-split manifests with content hashes, the frozen noise-mapping matrix, result tables, guard/regression tests, and table-

and figure-generation scripts—are released in a public repository, https://github.com/One-bit-007/graph-cold, and archived on Zenodo, https://doi.org/10.5281/zenodo.21873516.

Competing interests

The authors declare that they have no competing interests.

Funding

No specific funding was received for this work.

Authors' contributions

Tianqi Guo: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization. **Weiguo Wang**: Resources, Writing — review & editing. **Jian Wang**: Resources, Writing — review & editing. **Shiyin Zhu**: Resources, Writing — review & editing, Project administration.

Acknowledgements

Not applicable.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI assistants (Kimi, Moonshot AI; GPT- and Claude-based coding assistants) to support code development and debugging, experiment orchestration and execution of the pre-registered protocol, preliminary summarization of experimental outputs, and language editing and drafting of the manuscript and submission documents. All protocol decisions, statistical interpretations, and reported claims were independently reviewed and verified by the authors against the frozen experimental artifacts (tracked result files, audit reports, and regression tests), and the authors take full responsibility for the content of the publication.